

AI Design of Quantum Processors

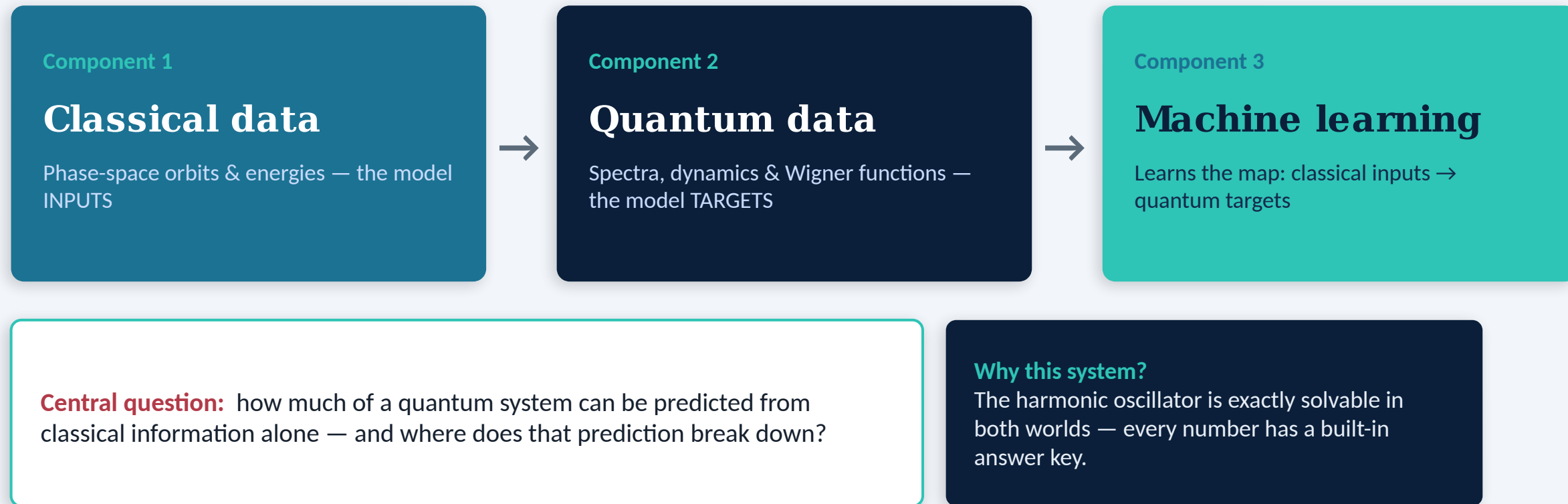
Components 1 & 2 — Generating classical and quantum data for the harmonic oscillator

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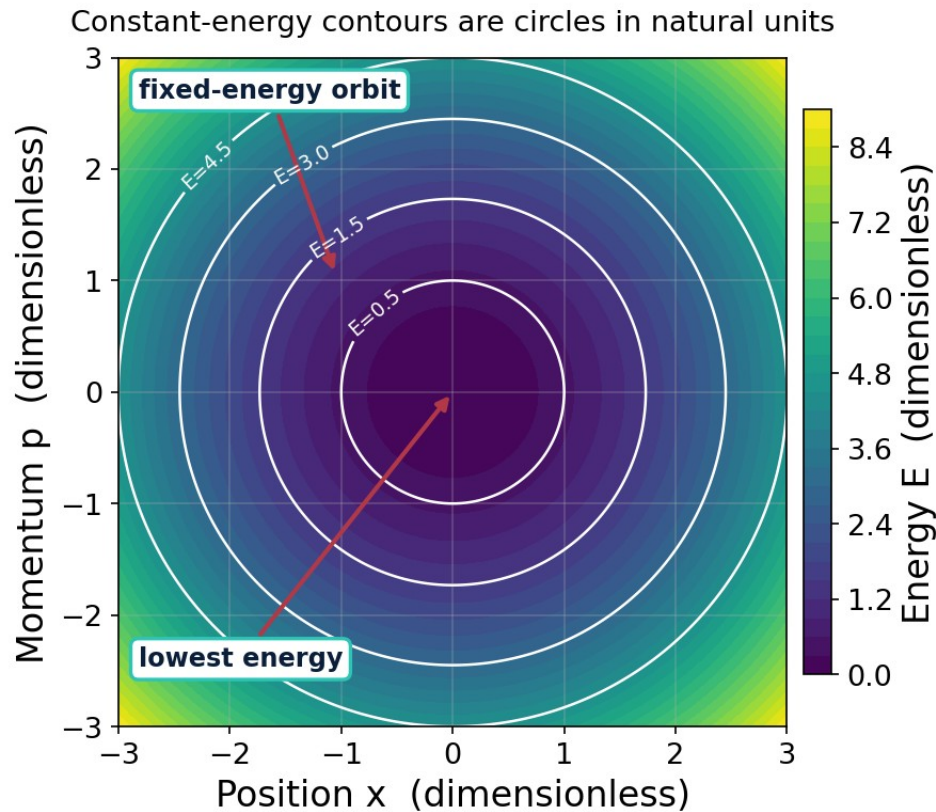
We test whether classical data can predict quantum properties, using the exactly-solvable harmonic oscillator.



Results

Classical (Component 1) and quantum (Component 2) data — verified against exact formulas.

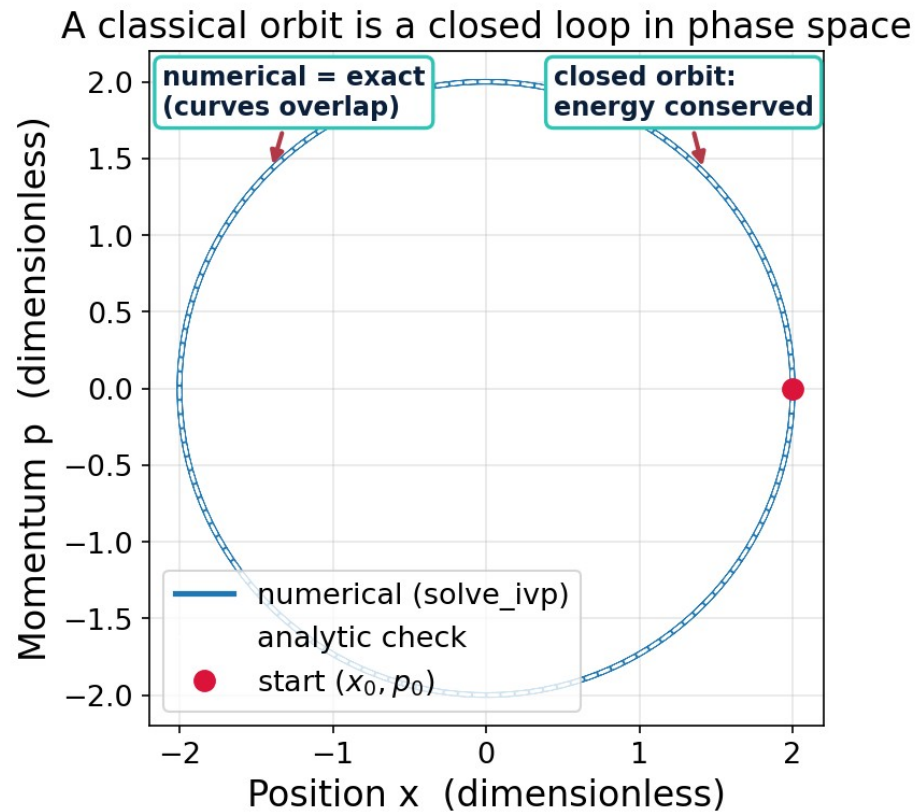
Constant-energy contours are closed elliptical orbits in phase space.



WHAT IT SHOWS

- Color is total energy $E(x,p)$; white rings are constant-energy curves.
- Each ring is one allowed classical orbit — the system is locked onto it.
- In natural units the ellipses become circles; larger energy = larger ring.
- This map defines the classical states the ML model will draw inputs from.

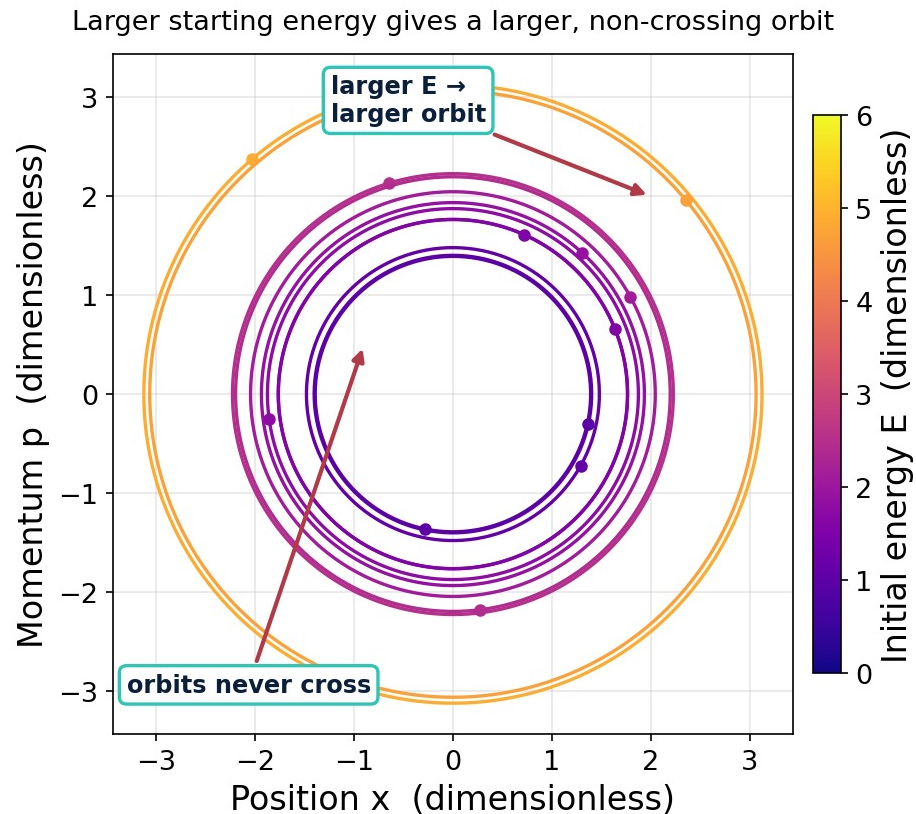
Numerical trajectories match the exact solution and conserve energy.



WHAT IT SHOWS

- `solve_ivp` steps Hamilton's equations forward in time to trace the orbit.
- Numerical path overlaps the analytic solution — the solver is verified.
- The orbit closes on itself and energy drift is only $\sim 1e-9$ — the solver conserves energy as it should.
- Confidence here transfers to systems with no exact answer later.

Higher energy gives larger, non-crossing orbits — the classical feature set.

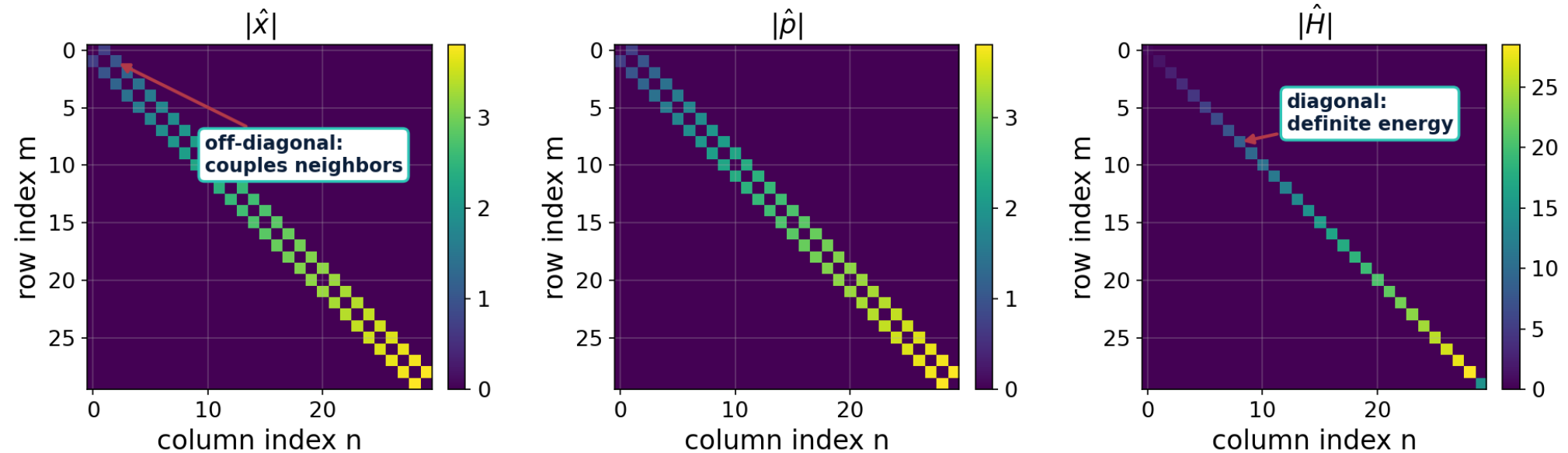


WHAT IT SHOWS

- Twelve random initial conditions, each colored by its conserved energy.
- Orbits never intersect: a unique state has a unique future.
- Saved as `classical_trajectories.npy` — the inputs for Component 3.
- Nested structure is exactly the 'feature space' ML will learn from.

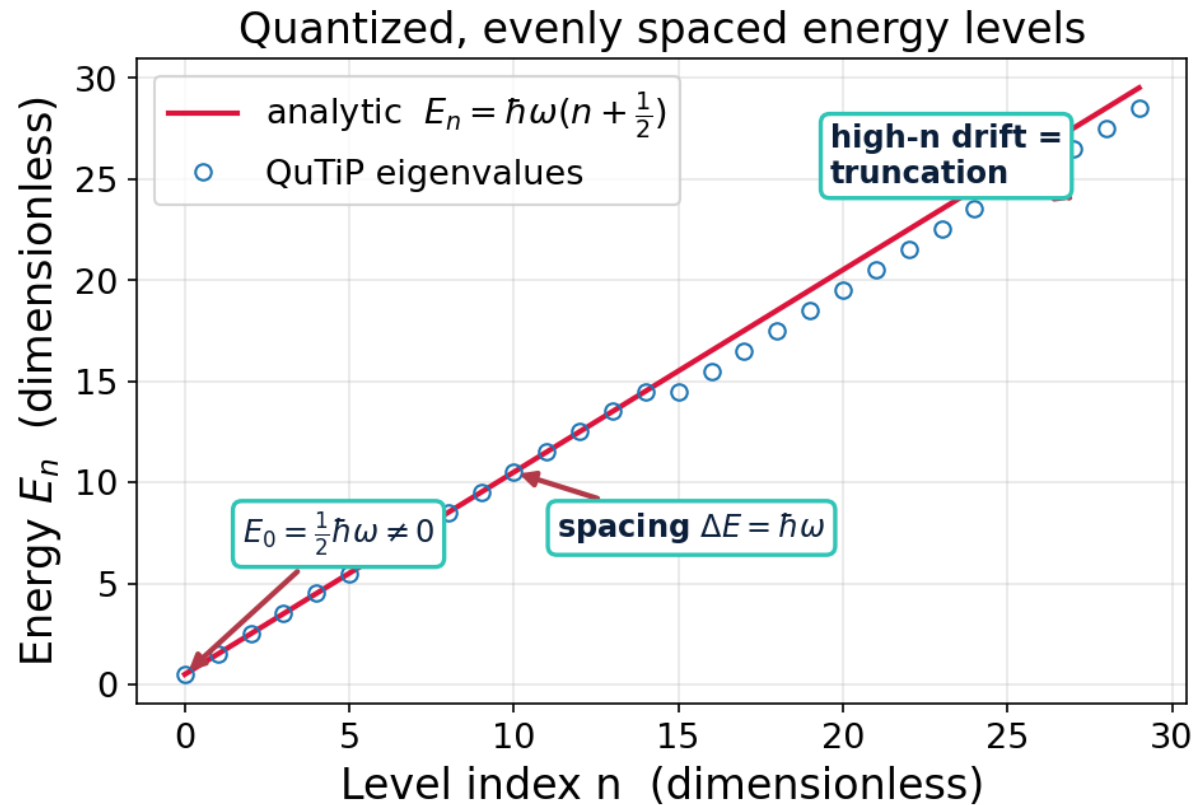
The quantum operators reveal the ladder structure of the oscillator.

Operator matrices in the energy (Fock) basis



Takeaway: $\hat{H}$ is diagonal (each level has a definite energy), while $\hat{x}$ and $\hat{p}$ have only off-diagonal entries — they connect a level to its immediate neighbors.

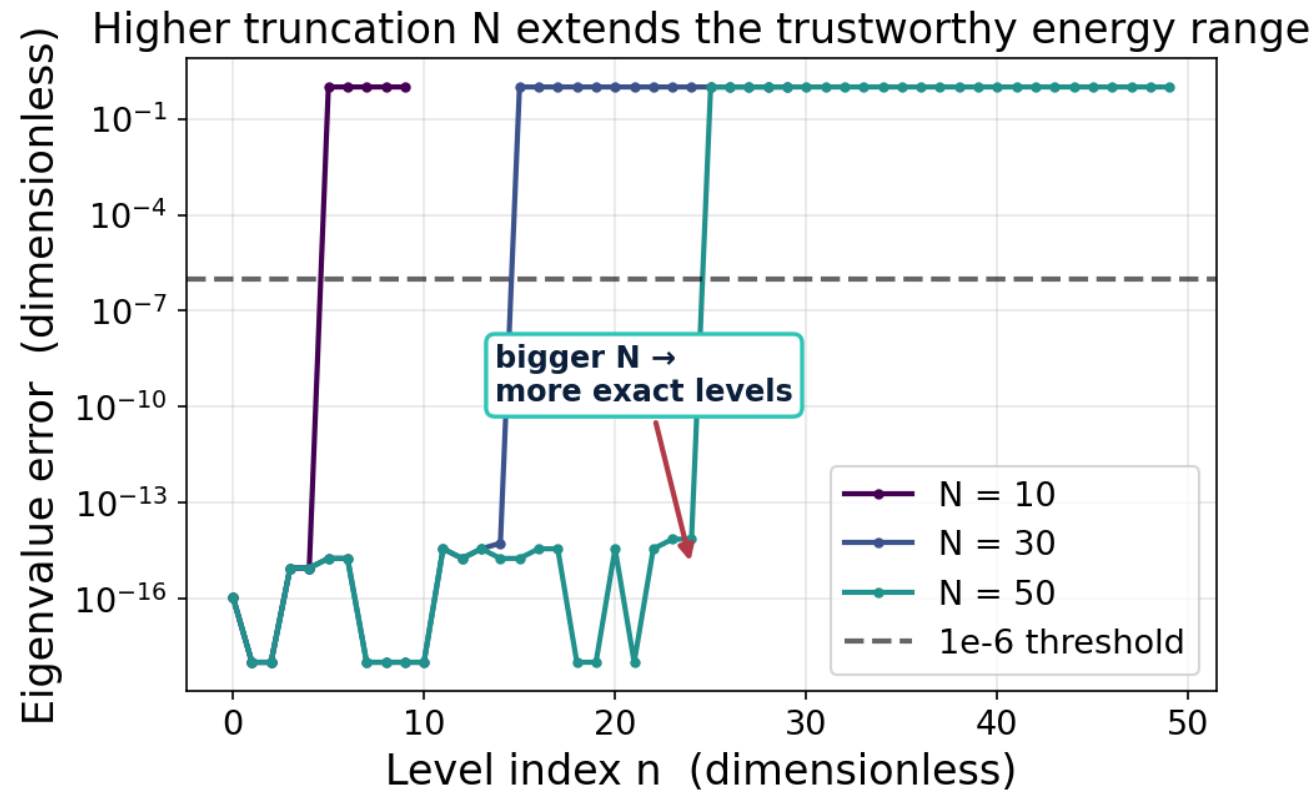
Quantum energy is discrete, evenly spaced, with a nonzero zero-point energy.



WHAT IT SHOWS

- QuTiP eigenvalues land exactly on the analytic line $E_n = \hbar\omega(n + \frac{1}{2})$.
- Ground state $E_0 = \frac{1}{2}\hbar\omega \neq 0$ — a purely quantum effect.
- Levels are equally spaced by $\hbar\omega$ (the ladder).
- High-n points drift: the expected signature of finite truncation.

The truncation level N sets how many energy levels are trustworthy.

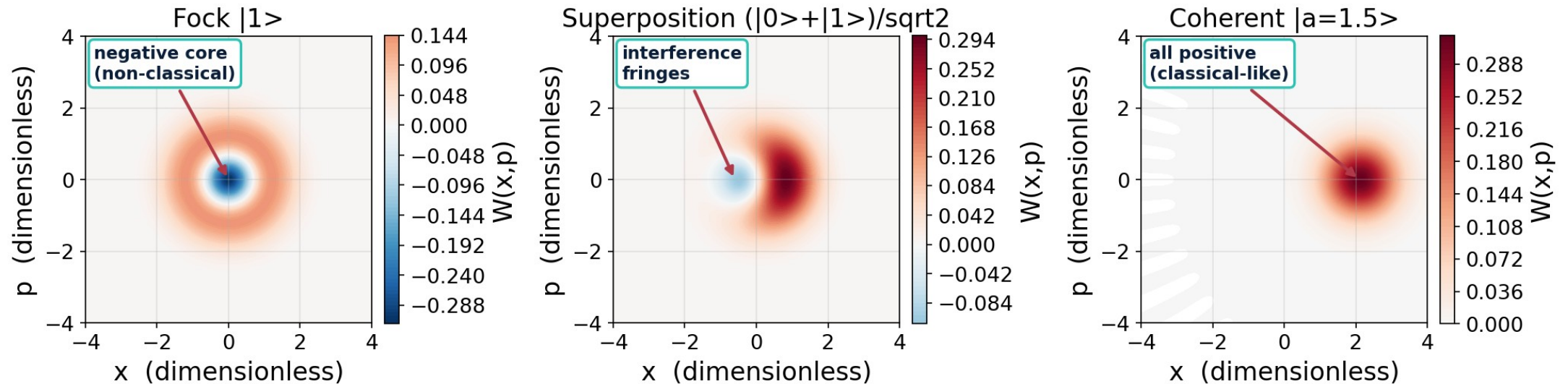


WHAT IT SHOWS

- Eigenvalue error stays at machine-zero for low levels, then blows up.
- Roughly the lowest $N/2$ levels are exact for a given truncation N .
- Raising N pushes the trustworthy range higher ($N=10 \rightarrow 5$, $30 \rightarrow 15$, $50 \rightarrow 25$).
- Rule: increase N until the quantity of interest stops changing.

Wigner functions expose non-classicality through negativity and interference.

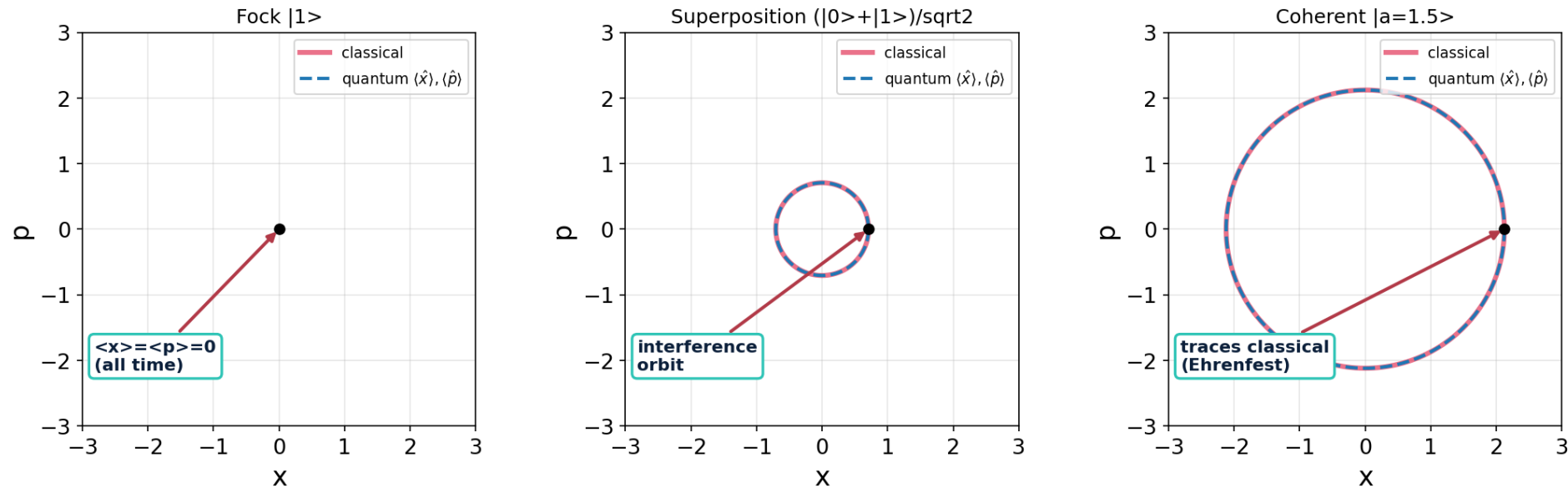
Wigner functions at $t = 0$ (blue = negative = non-classical)



Takeaway: The Fock state has a negative core and the superposition shows interference fringes (both impossible classically); the coherent state stays positive — the most classical state.

Quantum averages trace the classical orbit, but averaging hides the spread.

Quantum averages vs. classical orbits



Takeaway: For the coherent state $\langle \hat{x} \rangle, \langle \hat{p} \rangle$ follow the classical circle exactly (Ehrenfest's theorem); the Fock state sits at the origin. The averages discard the variance and interference the Wigner picture keeps.

Open questions & next steps

What I don't yet understand

- What classical features best predict quantum targets — raw trajectories, energies, or phase-space densities?
- How should the Wigner target be represented for ML (full grid vs. summary features)?
- How far up in energy does the classical→quantum prediction stay reliable?
- Where exactly does the classical picture break down (negativity, interference)?

What I'd like to do next

- Package Component 1 & 2 outputs into clean input/target pairs for Component 3.
- Sweep more states and energies to enlarge the dataset.
- Skim scikit-learn and pick a first model + validation plan.
- Discuss with PI which quantum property to predict first.